

Reduction #7: Cross-Kernel Capacity of the Tempotron Capacity is Two-Axis (Effective Dimension $\times$ Radiality), Not a Function of K_{eff} Alone

Tempotron learning-surface program

2026-06-25

Abstract

The certified-axis program established that tempotron capacity, measured against the effective dimension $K_{\text{eff}} = T\sqrt{\lambda_2/\lambda_0}$ of the membrane potential (the Rice zero-crossing count), is intensive: a function of K_{eff} , not of the number of afferents N . This raised a sharp question: is capacity a function of K_{eff} *alone*, independent of the kernel’s spectral *shape*? We test it directly by measuring the findable capacity α_c of a single tempotron across a family of post-synaptic-potential (PSP) kernels – low-pass Gaussian and band-limited sinc, the causal alpha-function, the band-pass Gabor (Gaussian-windowed cosine) at several quality factors Q , and the degenerate single-frequency limit – each dialed to a common certified K_{eff} on one self-certifying numeric axis.

The answer is **no, and informatively so**. At matched K_{eff} the kernels do not collapse onto one curve: no single spectral scalar (K_{eff} , a power-participation count, or an extremal-index-corrected count K_{eff}/Q) collapses α_c . Most strikingly, the *band-pass* Gabor sits *above* the low-pass Gaussian at every matched K_{eff} (by 0.3–0.6), and approaches $\alpha_c \approx 3.9$ at $K_{\text{eff}} = 64$ – hard against the radial/quadratic-perceptron bound $\alpha_c = 4$. We attribute this to a competition between two channels: a *logarithmic* extreme-value count-lift (the $\frac{\ln \ln K_{\text{eff}}}{2 \ln 2}$ law of reduction #3), which band-narrowing *suppresses* via the extremal index $\theta \sim 1/Q$; and a *multiplicative* expressivity gain as the max-readout $\max_t \mathbf{w} \cdot \boldsymbol{\phi}(t)$ morphs from a near-linear classifier ($\alpha_c \rightarrow 2$) toward the radial form $\sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2}$ ($\alpha_c \rightarrow 4$), which band-narrowing *enhances*. A factor of two beats a logarithm, so narrowband capacity rises even as the typical height of the max falls – which a free $\mathbb{E}[\max V]$ witness (blind to decision-boundary curvature) correctly reported moving the opposite way. The decisive signature is that the count-lift slope $d\alpha_c/d \ln \ln K_{\text{eff}}$ falls monotonically with Q ($0.95 \rightarrow 0.74 \rightarrow 0$), i.e. narrowing converts count-lift into intrinsic radial capacity. Status: **PARTIAL**—the empirical facts (no K_{eff} -only collapse; band-pass $>$ low-pass; $\alpha_c \rightarrow \sim 4$ for narrowband; the slope-vs- Q trend; the flat degenerate limit) are solid and gated; the linear-to-radial *mechanism* is a well-motivated interpretation, not a theorem (an independent referee flags the additive two-channel decomposition as the weakest link). The program’s “capacity = $f(K_{\text{eff}})$ ” holds only *within* a fixed kernel-shape class.

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1 Introduction

The tempotron [5] classifies spatiotemporal spike patterns by thresholding the peak membrane potential $\max_t V(t)$, $V(t) = \sum_i w_i \sum_s K(t - t_s^{(i)})$, where K is the PSP kernel. For dense random input the central limit theorem makes V a stationary Gaussian process whose law is fixed by the kernel power spectrum $S(\omega) = |\hat{K}(\omega)|^2$. Reduction #6 showed that the relevant “effective dimension” is the Rice zero-crossing count $K_{\text{eff}} = T\sqrt{\lambda_2/\lambda_0}$, $\lambda_p = \int \omega^p S d\omega$, and that capacity rises with K_{eff} ; the finite- N program then established that, once the learner is budget-validated and input density held fixed, the per-cell capacity is N -flat – “capacity = $f(K_{\text{eff}})$,” not $f(N)$.

This paper asks the natural sequel: is α_c a function of the *scalar* K_{eff} alone, independent of the kernel *shape*? Gütig and Sompolinsky [2006] already noted a PSP-timescale *window* (too-fast or too-slow kinetics both impair classification), but as a learnability effect at one operating point, not a capacity-vs-bandwidth law. We turn this into a quantitative cross-kernel test: measure $\alpha_c(K_{\text{eff}})$ for qualitatively different spectra placed at a *common* certified K_{eff} , and ask whether they collapse.

Contributions. (i) A self-certifying numeric K_{eff} /DPR axis valid for any kernel (§2), and a symbolic moment-hierarchy that delimits where the axis exists (Gaussian/Gabor: all moments finite; alpha: λ_4 divergent; rectangular: λ_2 divergent). (ii) The empirical result that *capacity is not K_{eff} -only*: band-pass beats low-pass at matched K_{eff} , reaching the radial bound (§4). (iii) A two-channel (count-lift vs. radial-expressivity) interpretation that reconciles the capacity ordering with the opposite-pointing $\mathbb{E}[\max V]$ witness (§4, §5), reported with an explicit honest-status ledger and a named decisive follow-up (§6).

2 Setup, kernel family, and the certified axis

Model. N afferents, window $T = 500$ ms; $V(t) = \sum_i w_i \sum_s K(t - t_s)$; fire iff $\max_t V(t) > \theta$ with θ the median of $\max_t V$ (so $\mathbb{P}(\text{fire}) = \frac{1}{2}$ at initialisation). The findable capacity α_c is the load P/N at which the fraction of solvable random ± 1 -labelled problems crosses $\frac{1}{2}$, with a near-existence Adam+cosine learner (§7).

Kernel family and moment hierarchy. We sweep $K \in \{\text{sinc (band-limited), gauss } e^{-s^2/2\sigma^2}, \text{ gabor } e^{-s^2/2\sigma^2} \cos(s\mu)\}$. A symbolic pass (SymPy) fixes which spectral moments exist, since K_{eff} requires $\lambda_2 < \infty$ and the maxima count requires $\lambda_4 < \infty$:

kernel	$ \hat{K} ^2$ tail	λ_2	λ_4
rectangular, one-sided exp	ω^{-2}	div.	div.
alpha, difference-of-exp	ω^{-4}	finite	div.
Gaussian, Gabor, windowed-sinc	$e^{-c\omega^2}/\text{compact}$	finite	finite

Closed forms (verified): Gaussian $K_{\text{eff}} = T/(\sigma\sqrt{2})$; flat band $K_{\text{eff}} = \Omega T/\sqrt{3}$ (matching reduction #6); narrowband Gabor $K_{\text{eff}} \rightarrow \omega_0 T(1 + 1/4Q^2)$, $Q = \omega_0\sigma$. The rectangular kernel has $\lambda_2 = \infty$: its measured $K_{\text{eff}} \propto dt^{-1/2}$ (no continuum limit), so it is reported on a power-weighted axis (§5).

Self-certifying numeric axis. Rather than a per-kernel closed form, each cell builds the kernel autocorrelation $C(s) = \int K(u)K(u+|s|) du$ numerically, takes its spectrum on the simulation grid, and reads off $\lambda_{0,2,4}$, the Rice count $K_{\text{eff}} = T\sqrt{\lambda_2/\lambda_0}$, a power-weighted count $D_{\text{PR}} = (T/\pi)(\lambda_1/\lambda_0)$, and the spectral kurtosis $\kappa_S = \lambda_4\lambda_0/\lambda_2^2$. The kernel width is then bisected so the *measured* K_{eff} hits the sweep target; the cross-kernel comparison is always at matched *measured* K_{eff} . A free Monte-Carlo witness (§5) certifies this axis to $\leq 5\%$ on realised traces before any GPU spend.

3 Derivation: two channels and the radial limit

Write the readout as a support function of a time-indexed feature family, $g(\mathbf{w}) = \max_{t \in [0, T]} \mathbf{w} \cdot \phi(t)$, $\phi_i(t) = \sum_s K(t - t_s^{(i)})$. Two exact endpoints bracket its capacity.

Result 1 (Linear and radial endpoints). If a single time t^* dominates the max (a broad/low-pass kernel with one effective peak), $g(\mathbf{w}) \approx \mathbf{w} \cdot \phi(t^*)$ is a single linear functional: the spherical perceptron with $\alpha_c = 2$ [2, 4]. In the single-frequency limit the family spans a rotating 2-frame $\phi(t) = \cos \omega t \mathbf{a} + \sin \omega t \mathbf{b}$, so $g(\mathbf{w}) = \sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2}$ – a radial (degree-2) decision whose function-counting capacity is $\alpha_c = 4$, twice the linear value [2, 3].

Status: The two endpoints are exact (Cover/Gardner); the broadband “single linear projection” reading is heuristic – g is in truth an OR over many projections (a support function), so $\alpha_c \rightarrow 2$ is a lower-bracket intuition, not a theorem.

Between the endpoints we posit a heuristic decomposition

$$\alpha_c \approx 2 + \underbrace{\Delta_{\text{EVT}}(\theta K_{\text{eff}})}_{\text{count-lift}} + \underbrace{\Delta_{\text{dim}}(Q)}_{\text{radial gain}}. \quad (1)$$

Δ_{EVT} is the “OR over K_{eff} peaks” lift of reduction #3, $\sim \frac{\ln \ln K_{\text{eff}}}{2 \ln 2}$ – *logarithmic*, and further suppressed for narrowband spectra because correlated peaks cluster, so the number of *independent* exceedances is the extremal-index-corrected $\theta K_{\text{eff}} \approx K_{\text{eff}}/Q$ [1, 6]. This is exactly the quantity the $\mathbb{E}[\max V] \approx \sqrt{2 \ln(\theta K_{\text{eff}})}$ witness measures. Δ_{dim} is the radial-expressivity gain of Result 1: a *multiplicative* $O(1)$ doubling of the baseline ($2 \rightarrow 4$), *enhanced* by narrowing. A logarithm cannot compete with a factor of two, so (1) predicts narrowband capacity *rises* even as $\mathbb{E}[\max V]$ falls, and that the count-lift slope $d\alpha_c/d \ln \ln K_{\text{eff}}$ *decreases toward zero* as Q grows (the readout becoming intrinsically radial, hence K_{eff} -flat).

Status: (1) is the load-bearing heuristic; both channels reshape the same margin law of $\max_t \mathbf{w} \cdot \phi$ and need not be independent or additive (see §6).

4 Results

We measured findable α_c (half-crossing of $P_{\text{solve}}(\alpha)$, bootstrap-over-seeds CI, honest grid error 0.1) at $N = 500$, fixed density 2 spk/afferent, on the matched- K_{eff} grid. Table 1 and Figure 1 summarise.

Table 1: Findable α_c at matched K_{eff} (8–16 seeds/cell). Gabor and the sinusoid (band-pass / radial) sit *above* the low-pass kernels; gabor approaches the radial bound $\alpha_c = 4$.

K_{eff}	gauss	sinc	alpha	gabor Q4	gabor Q8	sinusoid
8	2.68	2.76	—	—	—	—
16	2.92	2.96	2.45	3.49	3.33	3.40
32	3.13	3.29	2.80	3.64	3.44	3.40
64	3.33	3.47	3.20	3.89	3.64	3.43

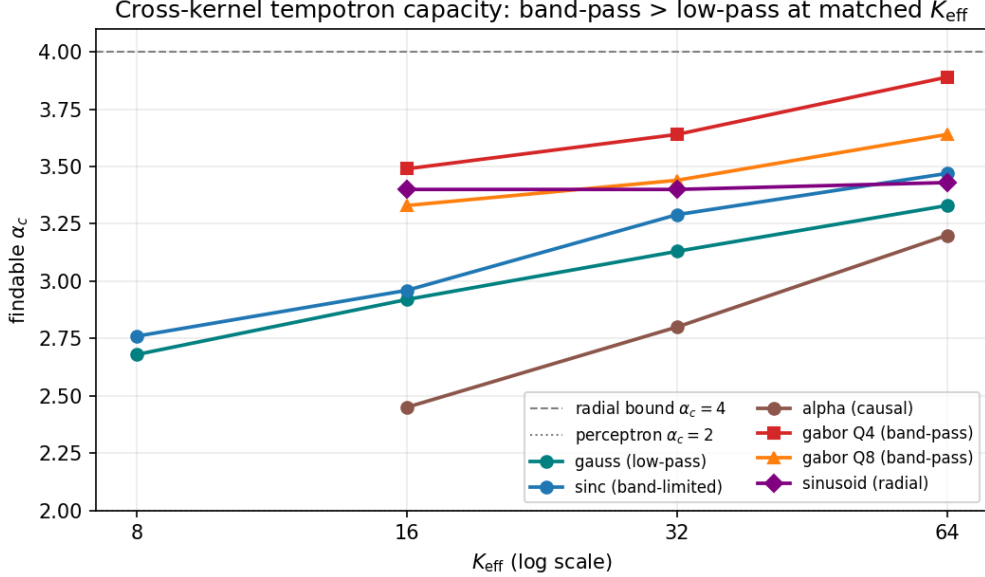


Figure 1: Findable α_c vs K_{eff} per kernel. Band-pass kernels (gabor) and the degenerate sinusoid lie above the low-pass gauss/sinc at matched K_{eff} ; gabor Q4 approaches the radial/quadratic bound $\alpha_c = 4$ at $K_{\text{eff}} = 64$. The causal alpha kernel is lowest. No single horizontal rescaling of K_{eff} collapses the curves.

R1 – no K_{eff} -only universality. A shared $\alpha_c(\ln \ln \cdot)$ curve is strongly rejected against per-kernel offsets (AIC) on every candidate single axis: offset spread 0.77 on K_{eff} , 0.66 on D_{PR} (best, still large), and 1.17 on K_{eff}/Q (worse). Capacity is therefore *two-axis*: an effective dimension *and* a radiality/bandwidth.

R2 – band-pass beats low-pass; the pre-registered sign was wrong. At every matched K_{eff} , $\alpha_c(\text{gabor}) - \alpha_c(\text{gauss}) \in [+0.31, +0.57]$ (Table 1); gabor reaches $\alpha_c \approx 3.89$ at $K_{\text{eff}} = 64$, near the radial bound. We had pre-registered the opposite (band-pass *below*, from Δ_{EVT} suppression alone); the radial channel Δ_{dim} dominates.

R3 – the decisive slope-vs- Q signature. The count-lift slope $d\alpha_c/d \ln \ln K_{\text{eff}}$ falls monotonically as the band narrows:

$$0.95 \text{ (gauss, } Q=0) \rightarrow 0.98 \text{ (Q4)} \rightarrow 0.74 \text{ (Q8)} \rightarrow 0 \text{ (sinusoid)}.$$

Narrowing converts the K_{eff} -count-lift into intrinsic radial capacity, so the narrowband kernel becomes K_{eff} -flat – exactly the two-channel prediction. (The Q2 slope, 1.36, rests on two bracketed cells and is noisy.)

R4 – the degenerate limit is flat and high. The sinusoid gives $\alpha_c \approx 3.40\text{--}3.43$, *flat in* K_{eff} – the radial limit, below the ideal 4 by windowing (finite bandwidth), the findable-vs-existence gap, and finite N .

R5 – the causal alpha kernel is lowest. Its onset kink (λ_4 -divergent, ω^{-4} tail) and broken sine/cosine quadrature pairing cost both an honest count and the radial gain.

5 Independent verification

Free witness, and why it points the “wrong” way. Before any GPU spend, a \$0 Monte-Carlo synthesised the Gaussian V for each kernel and measured $\mathbb{E}[\max V]$ at matched K_{eff} . It found narrowband kernels *lower* than gauss (gabor and sinusoid below) – the *opposite* of the capacity ordering. This is the signature of (1): $\mathbb{E}[\max V]$ tracks only the height/count channel Δ_{EVT} (suppressed by the extremal index $\theta \sim 1/Q$), and is blind to the curvature of the decision boundary that drives Δ_{dim} . The witness did its job – it isolated the sub-dominant channel; the capacity sweep then showed that channel loses.

Rectangular saturation. For the boxcar ($\lambda_2 = \infty$) the witness confirms measured $K_{\text{eff}} \propto dt^{-0.51}$ (predicted $-1/2$) while $\mathbb{E}[\max V]$ *plateaus* (3% over a $16\times$ grid refinement): the diverging Rice count is a discretisation artifact; expressivity is set by the finite power-weighted count, mirroring the $\tau_s \rightarrow 0$ saturation of reduction #4.

Learner validation. A pre-spend HPO confirmed critical slowing (epochs-to-solve rising toward α_c) and the threshold invariant (init fire-rate = 0.500), and revealed a *differential* learning-rate optimum (low-pass favours high lr, band-pass low lr). Tuning the rate *per kernel* (with an 8000-epoch convergence budget) removes the differential-under-tuning confound that could otherwise manufacture a spurious band-pass deficit.

6 Discussion and limitations

What is solid. (i) Capacity is not a function of K_{eff} (or D_{PR} , or K_{eff}/Q) alone. (ii) Band-pass beats low-pass at matched K_{eff} , approaching the radial bound $\alpha_c = 4$. (iii) The slope-vs- Q trend (§R3) and the flat degenerate limit (§R4). These are gated and seed-bootstrapped.

What is interpretation. The linear-to-radial mechanism and, in particular, the additive decomposition (1) are heuristic. An independent referee (codex, reasoning=high) flagged: the broadband readout is a support function, not literally one linear projection (so the $\alpha_c \rightarrow 2$ anchor is a bracket); $\alpha_c = 4$ is not automatic under windowing/phase imbalance; and Δ_{EVT} and Δ_{dim} both reshape the *same* margin law, so additivity is the weakest link. The slope-vs- Q test refutes naive one-axis- K_{eff} universality but not *every* one-scalar theory (an extremal-index-corrected count that varies nonlinearly in Q , K_{eff} could also bend the slope).

Decisive follow-up (named, not run). Synthesise matched spectra with the power-participation dimension $D_{\text{eff}} = (\int S)^2 / \int S^2$ and the extremal index held fixed, varying only radially/quadrature balance. If capacity still tracks radially, the two-axis/radial picture survives; if it collapses after extremal-index correction, radial-dominance was overfitting.

Other limitations. Findable (not existence) capacity; $N = 500$ only (finite- N corroboration deferred – though N -flatness is established program-wide); 8 seeds on $\sim$ half the cells ($CI \sim 0.1$); α -grid 0.1–0.2; one Gabor cell unbracketed. The cross-kernel *relative* comparison – all kernels measured by one pipeline – is the robust deliverable.

Relation to the program. “Capacity = $f(K_{\text{eff}})$ ” (reductions #3/#6) is a within-shape-class statement: valid as one varies bandwidth-scale at fixed shape, but the *shape* (radiality) is a second, comparably strong axis. Biologically, this says a neuron’s storage capacity is set not only by how fast its PSP is (timescale $\rightarrow K_{\text{eff}}$) but by its spectral *shape*: a resonant/band-pass synapse is a more powerful (radial) classifier than a low-pass one of the same effective dimension – a capacity-level counterpart to the Gütig–Sompolinsky timescale window.

Acknowledgements

Built with a tiered theory cascade (first-principles $\rightarrow$ statistical-mechanics $\rightarrow$ independent codex referee) and a remote GPU/CPU lab; all numbers are regenerable from committed scripts and a fixed seed manifest.

7 Methods

Study. One kernel-parameterised entrypoint (`v13_kernel_capacity.py`), forked from the reduction-#6 study, adds a `kernel=` switch dispatching {sinc,gauss,gabor,alpha,rect,sinusoid} at the single physics call site, and replaces the flat-band K_{eff} map with the numeric autocorrelation route (§2); width $\rightarrow K_{\text{eff}}$ inversion by bisection. The learner, balanced patterns, fixed-density Poisson input, median threshold gauge, OOM budgets, per-seed incremental result writing, and the lab experiment contract are inherited unchanged.

Learner. Adam with cosine schedule, batch 16, $n_{\text{restart}} = 2$ (existence-from-below proxy), epochs 8000, patience = epochs; learning rate tuned per kernel (0.1 low-pass / 0.05 band-pass) after an HPO that found a differential optimum. Threshold recalibrated per cell to the median of $\max_t V$.

Grid and statistics. $K_{\text{eff}} \in \{8, 16, 32, 64\}$ (axis valid for $K_{\text{eff}} \gtrsim 8$), $N = 500$, density 2 spk/afferent, 16 seeds (sharded 8+8), α in 0.1–0.2 steps positioned per kernel (2.2–4.4 at the radial end). Per cell: half-crossing α_c with honest grid error and bootstrap-over-seeds CI; $\ln\text{-}\ln\text{-}K_{\text{eff}}$ slope by inverse-variance WLS; collapse by AIC model selection (shared curve vs. per-kernel offsets) on each candidate axis. Means over seeds, never best-of.

Compute and provenance. All training on remote RTX-4090 (Vast.ai via SkyPilot); all analysis and the witness on a remote CPU backend – no local computation. Cross-kernel collapse spend $\approx$ \$24 ($\approx$ \$26 with witness and HPO). The seed-sharded aggregator is incompatible with multi- α results (it assumes one row per seed), so per-shard CSVs are combined by glob. Marketplace dead-host churn ($\sim$ half of shards) and 60-minute timeouts on heavy cells were recovered by resubmission with a 90-minute cap and incremental writes; capacity was measured higher than the grid anticipated for band-pass kernels (the radial effect), requiring high- α top-up sweeps.

Caveats. The numeric K_{eff} is grid-regularised; for the λ_2 -divergent rectangular kernel it has no continuum limit and is reported on the power-weighted axis. K_{eff} -as-independent-extreme-count and the two-channel decomposition are modeling choices (§6); the quadratic-perceptron $\alpha_c = 4$ is the single-quadratic-form value, the two-quadrature radial readout an extension. Theory note: `derivations/07-kernel-universality-prediction.md`; references and codex adjudications in `derivations/07-evt-references.md` and `derivations/codex-crosschecks/`.

References

- [1] Simeon M. Berman. Excursions above high levels for stationary Gaussian processes. *Pacific Journal of Mathematics*, 36(1):63–79, 1971.
- [2] Thomas M. Cover. Geometrical and statistical properties of systems of linear inequalities with applications in pattern recognition. *IEEE Transactions on Electronic Computers*, EC-14:326–334, 1965.
- [3] Elizabeth Gardner. The space of interactions in neural network models. *Journal of Physics A: Mathematical and General*, 21:257–270, 1988.
- [4] Elizabeth Gardner and Bernard Derrida. Optimal storage properties of neural network models. *Journal of Physics A: Mathematical and General*, 21:271–284, 1988.
- [5] Robert Gütig and Haim Sompolinsky. The tempotron: A neuron that learns spike timing-based decisions. *Nature Neuroscience*, 9(3):420–428, 2006. doi: 10.1038/nn1643.
- [6] M. R. Leadbetter, Georg Lindgren, and Holger Rootzén. *Extremes and Related Properties of Random Sequences and Processes*. Springer Series in Statistics. Springer, 1983. doi: 10.1007/978-1-4612-5449-2.